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# Impact of Social Vulnerability Index and Distance to Tertiary Ocular Emergency Department in Patients Presenting with Open Globe Injury

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***Purpose:*** To evaluate the impact of Social Vulnerability Index (SVI) and the distance to a tertiary eye emergency department (ED) on presenting visual acuity (VA), final VA, and rates of VA change over time in patients presenting with open globe injury (OGI).

***Design:*** Retrospective cohort study.

***Participants:*** Patients with OGI from 2018 to 2021 were identified from the electronic health records using the International Classification of Diseases coding system.

***Methods:*** Data collected included demographics, insurance status, follow-up visits, and VA measurements. Social Vulnerability Index scores were calculated using a US Census Geocoder and stratified into quartiles, and distances to the eye ED were calculated. Simple logistic regression models evaluated the associations of presenting and final VAs with SVI, distance, baseline characteristics, and time from injury to presentation, adjusting for potential confounders. Univariable and multivariable linear regression models analyzed the associations between VA rates of change over time and SVI, distance to ED, demographics, insurance, and presenting VA.

***Main Outcome Measures:*** The relationship between SVI and distance on presenting VA, final VA, and rates of longitudinal VA change over time.

***Results:*** A total of 446 patients presented with an OGI. Of these patients, 337 (75.0%) with complete SVI data were included in the final analysis. Social Vulnerability Index was associated with nearly a fourfold increased risk of worse presenting VA in populations with higher vulnerability compared with those with lower overall vulnerability (odds ratio [OR] = 3.85; 95% confidence interval [CI]: 1.34–11.05; $P = 0.012$). Greater distance from the eye ED was also a contributing factor to higher risk of poorer presenting VA (OR = 2.55; 95% CI: 1.42–4.60; $P = 0.002$). On multivariable analyses, longitudinal VA change over time was not impacted by SVI ($P > 0.05$ for all groups), and final visual outcome was not associated with either SVI or distance to the eye ED.

***Conclusions:*** Greater distances from the eye ED and social vulnerability were associated with a higher risk of worse VA at presentation. However, no significant impact of SVI or distance was observed on long-term VA changes and final visual outcome on the multivariable analyses.

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*Supplemental material available at www.ophthalmologyscience.org.*

Open globe injury (OGI) is an ocular emergency resulting in 1.6 million cases of blindness annually, making it the leading cause of monocular blindness in the world.[1] The prognosis and management of these injuries depend on several factors, with presenting visual acuity (VA) being one of the most important predictor of visual outcomes.[2–6]

Although clinical factors such as VA and retinal detachment have been demonstrated to affect the likelihood of visual recovery, socioeconomic status can affect both the incidence and outcomes of OGI.[7] For example, individuals with lower socioeconomic status, as defined by income, educational level, and employment status, are at higher risk of sustaining these injuries.[7] In addition, sociodemographic characteristics, such as race and ethnicity, have been found to be an independent prognostic factor for a poor visual outcome.[6] The influence of social factors on ocular trauma highlights the need to better understand this relationship and address disparities in care for these populations.

The Social Vulnerability Index (SVI) is a valuable tool for identifying at-risk populations and guiding interventions to reduce the incidence and improve the outcomes of OGI. The SVI is a tool developed by the Centers for Disease

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Control to identify populations susceptible to hazards based on 15 social census variables, grouped into the following 4 subthemes: socioeconomic status, household composition and disability, minority status and language, and housing and transportation.[8] Each factor is graded from 0 to 1, with higher values indicating higher vulnerability. A recent study found that patients with greater SVI levels were associated with worse ocular trauma score.[9] Previous studies have also explored the relationship between SVI and visual outcomes in the setting of glaucoma and found that patients in areas of greater social vulnerability face an elevated risk of poor visual prognosis, delays in receiving glaucoma surgical intervention, and lower adherence to eye care.[10–12]

It is not yet fully understood if SVI can affect visual recovery in patients who experience OGI. Patients who sustain OGI require longitudinal follow-up to monitor and manage secondary complications, such as glaucoma, cataracts, and retinopathy, that can significantly influence visual outcomes.[5] Furthermore, studies have shown that patients with greater social vulnerability are less likely to adhere to ophthalmology follow-up care.[12] These findings highlight the need to examine how social vulnerability can affect access to care and visual outcomes in patients with ocular trauma.

This study aimed to assess the influence of SVI and the distance to a tertiary ocular emergency center on initial and final VA, as well as their impact on the longitudinal rate of change in VA over time in patients with OGI.

## Methods

This study was approved by the University of Miami Institutional Review Board and adhered to the Declaration of Helsinki and the Health Insurance Portability and Accountability Act. A waiver of informed consent was granted due to the retrospective nature of the study.

### Inclusion and Exclusion Criteria

Patients of all ages were included based on International Classification of Diseases codes for ocular laceration or rupture (Appendix 1, available at www.ophthalmologyscience.org) between 2018 and 2021 in the electronic health records of Bascom Palmer Eye Institute emergency department (ED). These records were then manually reviewed to ensure strict inclusion of cases with a confirmed OGI diagnosis, as documented in the chart notes. The exclusion criteria included patients missing the required data to be successfully geocoded with SVI metrics or those lacking presenting VA. The collected data included demographics, health insurance status at presentation, mechanism of injury, date of injury, date of presentation to the ED, all visits from the initial ED presentation to the final follow-up visit, clinical examination findings at each visit, longitudinal data tracking changes in VA over time, and whether enucleation was performed. For patients with bilateral OGIs, the right eye was arbitrarily selected for analysis to ensure that only 1 eye per patient was included in the study for all patients in our cohort.

Patients' VAs were transformed into logarithm of the minimum angle of resolution (logMAR). The VA per visit was selected based on the best corrected available VA in the following sequence: (1) with correction, (2) pinhole, or (3) without correction, and used for analysis. Patient addresses were geocoded based on the US Census Geocoder. For insurance status, patients were classified based on the following presenting insurance status: uninsured (self-pay), public insurance (Medicare, Medicaid, Tricare), and private insurance (Blue Cross Blue Shield, HUMANA, Commercial). Furthermore, we classified patients based on their most frequently used insurance type across all visits during the follow-up period. If a patient had workers' compensation coverage at any point during the follow-up, they were classified under workers' compensation status.

### SVI and Distance Calculation

Data used to calculate the SVI were extracted from the electronic health record. The SVI overall theme and 4 subthemes were estimated by the US Census Geocoder and then stratified into quartiles to classify patients based on severity across varying levels of social vulnerability, in alignment with prior studies.[8] The quartiles were defined as follows: 0 to 25 indicating mild vulnerability, 25 to 50 representing moderate vulnerability, 50 to 75 denoting high vulnerability, and 75 to 100 signifying very high vulnerability. The overall theme, which combines all 15 variables from the 4 subthemes into a single score, was used to represent the social vulnerability of a census tract or county in our regression models.

Distances from patients' home addresses to Bascom Palmer Eye Institute in Miami, Florida, were calculated using Quantum Geographic Information System (version 3.10). The coordinates were initially georeferenced (the US Census Geocoder) using latitude and longitude and then transformed into Cartesian coordinates to compute the distances in feet. To enhance clarity, the distances were subsequently converted into miles. Distance was categorized into 2 groups: $>30$ miles and $\leq 30$ miles, to estimate a driving time of 30 to 60 minutes.

### Progression Outcomes and Statistical Analysis

Descriptive statistics were used to evaluate demographics and baseline characteristics of patients, including the calculation of means and standard deviations for normally distributed continuous variables and median with first and third quartiles for non normally distributed variables. The distribution of continuous variables was assessed using histograms and the Shapiro–Wilk test to evaluate normality. For comparisons between groups, independent samples *t* tests were conducted to assess differences in continuous variables with approximately normal distributions, and chi-square tests for categorical variables.

The associations between presenting VA, SVI, distance, and baseline demographics were examined. Visual acuity at presentation was categorized as a binary outcome, using 20/200 (logMAR 1.0) as the threshold for monocular blindness. Logistic regression models with robust sandwich estimations were applied to assess the relationships between eye-level variables at presentation, SVI, and distance. A multivariable model was used to adjust for sex, race, ethnicity, age, distance, time from injury to presentation date, and insurance status at presentation. The same analysis was conducted to evaluate the impact of SVI and demographic characteristics on the final recorded VA during follow-up, with the only modification being in the multivariable analysis additionally adjusting for presenting VA (categorized based on the latest World Health Organization classification,[13] with acuity worse than 20/40 categorized as mild, 20/60 as moderate, and 20/200 as severe), as well as insurance status being the most used insurance throughout follow-up.

The longitudinal rates of change in logMAR VA over time were assessed using linear mixed models with random effects placed at the eye level using random slopes and best linear unbiased

predictions. In short, best linear unbiased predictions assess individual eye slopes, yielding more precise estimates than ordinary least square linear regressions. A similar methodology for calculating longitudinal changes in VA and slopes has been utilized in previous studies.[14] Although survival analysis was initially considered, its feasibility was limited because most patients in our OGI cohort had already experienced the event of interest at baseline (i.e., VA worse than 20/200), leading to early censoring of many eyes and a small residual sample size. Consequently, we adopted an alternative approach by analyzing longitudinal rates of VA change across multiple follow-up visits to explore potential associations between the slope of change and factors such as SVI, distance, and demographics. A positive slope indicated worsening VA over time (i.e., no recovery), as VA is measured in logMAR (in which higher values represent poorer vision), with faster slopes signifying more rapid decline. Conversely, a negative slope indicated improving VA over time. Univariable and multivariable ordinary least squares linear regression models were employed to analyze associations between slope of VA change over time and factors including SVI, distance from the eye ED, baseline demographic characteristics (age, sex, and race), most used insurance throughout follow-up, and presenting VA based on the World Health Organization classification.[13]

Statistical analyses were performed with Stata 17 (Statacorp LP). Alpha level was set at 0.05 for all analyses.

## Results

A total of 446 patients presented with an OGI between 2018 and 2021. Of these, 337 (76%) with SVI data were included in the final analysis. The average follow-up time was 1.38 ± 1.49 years. Seventy-four patients (22.0%) self-identified as female and 86 (25.5%) reported their race as Black or African American. At presentation, most patients were uninsured (81.9%), although some had private (12.1%) or public (3.0%) insurance, 1.5% had workers' compensation, and 1.5% had unspecified status. However, when looking at the most frequently used insurance type across all visits during follow-up, 51.9% of patients had private insurance, 15.1% were covered by public insurance, 8.3% were under workers' compensation, 22.3% were uninsured, and 2.4% remained unspecified. The median distance for all patients was 19.3 miles (interquartile range: 8.6−63.4 miles). Of these, 208 (61.7%) patients lived within 30 miles of Bascom Palmer Eye Institute, while 129 (38.3%) patients were located >30 miles away. Among the cases, two-thirds of eyes (225, 67.0%) had a presenting VA worse than 20/200. This group were significantly older (51.5 ± 22.7 years) compared with those with a presenting VA better than 20/200 (41.2 ± 19.6 years, $P < 0.001$). In our cohort, average rate of longitudinal VA change over time was −0.32 ± 0.25 logMAR units per year overall, with approximately half of the patients improving at the last follow-up (179 patients, 53.1%). Most patients in our cohort (75.4%) presented to our eye center within 24 hours of injury, and 10.1% presented after >72 hours. For the overall SVI theme, most patients in our cohort were in the very high vulnerability category (54.3%), followed by high (24.0%), moderate (13.7%), and low (8.0%) categories. Twelve patients underwent enucleation, with the majority (75.0%) falling into the high and very high vulnerability subcategories. Detailed information on demographics and baseline characteristics are in Table 1.

Table 2 reports the univariable and multivariable models evaluating the correlation between presenting VA, demographics, overall, SVI theme, distance to the eye ED, and time from injury to presentation. In the univariable analysis, a distance >30 miles was significantly associated with approximately twice the likelihood of worse presenting VA compared with shorter distances (odds ratio [OR] = 1.89; 95% confidence interval [CI]: 1.16−3.09; $P = 0.010$). Other factors increasing the risk of worse presenting VA included female sex (OR = 2.08; 95% CI: 1.13−3.83; $P = 0.018$), Black race (OR = 1.93; 95% CI: 1.09−3.40; $P = 0.023$), as well as baseline age (OR = 1.24 per decade older; 95% CI: 1.12−1.38; $P < 0.001$). The SVI overall index was not statistically significantly associated with VA at presentation ($P > 0.05$ in all groups). The multivariable analysis assessing the impact of SVI vulnerability quartiles on presenting VA after OGI was adjusted for sex, race, ethnicity, age, insurance status at presentation, distance to the eye ED, and time from injury to presentation. The analysis yielded a significant association between worse presenting VA and overall SVI, as the high vulnerability category was associated with nearly a fourfold higher risk of monocular blindness at presentation (OR = 3.85 compared with the low vulnerability subclass, 95% CI: 1.34−11.05; $P = 0.012$). Distance to the eye ED remained a significant factor influencing worse presenting VA, with distances >30 miles associated with worse outcomes compared with those under this threshold (OR = 2.55; 95% CI: 1.42−4.60; $P = 0.002$). Black race also remained significantly associated with worse VA at baseline compared with non-Black race (OR = 4.03; 95% CI: 1.81−8.97; $P = 0.001$), along with baseline age (OR = 1.31; 95% CI: 1.16−1.49; $P < 0.001$).

Table 3 reports the univariable and multivariable models evaluating the correlation between slope of VA change over time, demographics, overall SVI theme, distance to eye center, presenting VA severity, and time to injury to presentation. In the univariable analysis, SVI theme did not show any significant associations with VA change over time ($P > 0.05$ for all). Distance from the eye ED was also not statistically significantly associated with the rate of VA change over time ($P = 0.600$). Race was a significant predictor, with Black patients having a slightly greater deterioration in VA compared with non-Black patients (ß = 0.076; 95% CI: 0.020−0.132; $P = 0.008$). Baseline age was also significantly associated with worsening VA (ß = 0.020 per decade older; 95% CI: 0.008−0.032; $P = 0.001$). In the multivariable analysis, adjusting for potential confounders, race remained a significant predictor of VA decline, with Black patients showing a greater slope of VA worsening in logMAR compared with non-Black patients (ß = 0.103, 95% CI: 0.025−0.181; $P = 0.010$). Baseline age continued to be significantly associated with a worse VA trajectory (ß = 0.025 per decade older; 95% CI: 0.011−0.040; $P < 0.001$).

Table 4 presents the univariable and multivariable models evaluating risk factors associated with poor final

Table 1. Demographics and Baseline Characteristics (N = 337)

| Characteristic | Value |
|---|---|
| Sex | |
| Male | 263 (78.0) |
| Female | 74 (22.0) |
| Race | |
| Non-Black | 236 (70.0) |
| Black | 86 (25.5) |
| Unspecified | 15 (4.5) |
| Ethnicity | |
| Non-Hispanic or Latino | 165 (48.9) |
| Hispanic or Latino | 166 (49.3) |
| Unspecified | 6 (1.8) |
| Baseline age, yrs | |
| Mean ± SD | 48.1 ± 22.3 |
| Follow-up, yrs | |
| Mean ± SD | 1.38 ± 1.49 |
| Time from injury to presentation | |
| Within 24 hrs | 254 (75.4) |
| Within 48 hrs | 14 (4.2) |
| Within 72 hrs | 13 (3.9) |
| >72 hrs | 34 (10.1) |
| Unspecified | 22 (6.5) |
| Insurance status at presentation | |
| Uninsured (self-pay) | 276 (81.9) |
| Private (BCBS, Commercial) | 41 (12.1) |
| Public (Medicare, Medicaid, Tricare) | 10 (3.0) |
| Workers' compensation | 5 (1.5) |
| Unspecified | 5 (1.5) |
| Insurance status most used during follow-up visits | |
| Uninsured (self-pay) | 75 (22.3) |
| Private (BCBS, Commercial) | 175 (51.9) |
| Public (Medicare, Medicaid, Tricare) | 51 (15.1) |
| Workers' compensation | 28 (8.3) |
| Unspecified | 8 (2.4) |
| Distance | |
| <30 miles | 208 (61.7) |
| >30 miles | 129 (38.3) |
| Distance, miles | |
| Mean ± SD | 73.2 ± 238.2 |
| Median [IQR] | 19.3 [8.6; 63.4] |
| Rate of VA change over time, logMAR units per year | −0.32 ± 0.25 |
| Mean ± SD | |
| SVI theme overall | |
| Low vulnerability (0%–25%) | 27 (8.0) |
| Moderate vulnerability (25%–50%) | 46 (13.7) |
| High vulnerability (50%–75%) | 81 (24.0) |
| Very high vulnerability (75%–100%) | 183 (54.3) |
| SVI subtheme 1 | |
| Low vulnerability (0%–25%) | 24 (7.1) |
| Moderate vulnerability (25%–50%) | 43 (12.8) |
| High vulnerability (50%–75%) | 68 (20.2) |
| Very high vulnerability (75%–100%) | 202 (59.9) |
| SVI subtheme 2 | |
| Low vulnerability (0%–25%) | 25 (7.4) |
| Moderate vulnerability (25%–50%) | 70 (20.8) |
| High vulnerability (50%–75%) | 105 (31.2) |
| Very high vulnerability (75%–100%) | 137 (40.6) |
| SVI subtheme 3 | |
| Low vulnerability (0%–25%) | 21 (6.2) |
| Moderate vulnerability (25%–50%) | 23 (6.8) |
| High vulnerability (50%–75%) | 65 (19.3) |
| Very high vulnerability (75%–100%) | 228 (67.7) |
| SVI subtheme 4 | |
| Low vulnerability (0%–25%) | 58 (17.2) |
| Moderate vulnerability (25%–50%) | 64 (19.0) |
| High vulnerability (50%–75%) | 102 (30.3) |
| Very high vulnerability (75%–100%) | 113 (33.5) |

BCBS = Blue Cross Blue Shield; IQR = interquartile range; logMAR = logarithm of the minimum angle of resolution; n = number; SD = standard deviation; SVI = Social Vulnerability Index; VA = visual acuity.
All values are n (%) unless otherwise noted.

VA after OGI. In the univariable analysis, a distance >30 miles was significantly associated with a higher likelihood of worse final VA (OR = 1.81; 95% CI: 1.16–2.83; $P$ = 0.009), as was female sex (OR = 2.00; 95% CI: 1.18–3.38; $P$ = 0.010), Black race (OR = 1.94; 95% CI: 1.17–3.21; $P$ = 0.010), and baseline age (OR = 1.39 per decade older; 95% CI: 1.24–1.57; $P$ < 0.001). Hispanic or Latino ethnicity was associated with a lower likelihood of worse final VA (OR = 0.51; 95% CI: 0.33–0.79; $P$ = 0.003). Additionally, a severe presenting VA was a strong predictor of poor final VA (OR = 22.62; 95% CI: 5.19–98.46; $P$ < 0.001). Public insurance status had a worse VA compared with self-uninsured patients (OR = 3.17; 95% CI: 1.50–6.69; $P$ =0.002). In the multivariable analysis, baseline age and worse presenting VA severity remained statistically associated with higher odds of poor final VA compared with those with mild severity (OR = 1.36 per decade older; 95% CI: 1.18–1.57; $P$ < 0.001; OR = 17.54; 95% CI: 3.43–89.83; $P$ = 0.001, respectively). The SVI overall theme and insurance status did not have any significant impact on final VA outcomes ($P$ > 0.05 in all categories).

## Discussion

Our study examined the influence of the SVI and geographic distance on OGI severity, assessed through presenting and final VAs, and examined the effect of SVI on longitudinal changes in VA over time. Higher SVI and greater distance to the tertiary ocular center were associated with worse presenting VA, but neither influenced VA progression nor final outcomes during follow-up in the multivariable analyses adjusting for potential confounders.

The presenting severity after OGI can be viewed multidimensionally and was influenced by both social vulnerability as well as the patient's distance to our emergency center. Our findings align with previous reports that demonstrate patients with greater social vulnerability are associated with worse ocular trauma score on presentation.[9] In our study, we found that patients from communities of high SVI are nearly 4 times more likely to present with monocular blindness compared with counties with low vulnerability (OR = 3.85; $P$ = 0.012). In a previous study by Schulz et al,[15] higher social vulnerability and more distressed communities as assessed by the Distressed

Table 2. Univariable and Multivariable Analysis Assessing Risk Factors for Poor Presenting Visual Acuity (Worse than 20/200) after Open Globe Injury

| Characteristic | Univariable Analysis | | Multivariable Analysis | |
|---|---|---|---|---|
| | *OR (95% CI)* | P *Value** | *OR (95% CI)* | P *Value** |
| SVI theme overall | | | | |
| Low vulnerability (0%–25%) | Base | | Base | |
| Moderate vulnerability (25%–50%) | 2.03 (0.75–5.50) | 0.163 | 2.40 (0.80–7.19) | 0.117 |
| High vulnerability (50%–75%) | 2.29 (0.92–5.67) | 0.075 | 3.85 (1.34–11.05) | **0.012** |
| Very high vulnerability (75%–100%) | 1.42 (0.63–3.21) | 0.403 | 2.21 (0.79–6.15) | 0.131 |
| Distance | | | | |
| >30 miles | 1.89 (1.16–3.09) | **0.010** | 2.55 (1.42–4.60) | **0.002** |
| Sex | | | | |
| Female | 2.08 (1.13–3.83) | **0.018** | 1.81 (0.89–3.70) | 0.103 |
| Race | | **0.023** | | **0.001** |
| Non-Black | Base | | Base | |
| Black | 1.93 (1.09–3.40) | | 4.03 (1.81–8.97) | |
| Ethnicity | | 0.260 | | 0.084 |
| Non-Hispanic or Latino | Base | | Base | |
| Hispanic or Latino | 0.77 (0.49–1.22) | | 1.78 (0.93–3.41) | |
| Baseline age per decade older | 1.24 (1.12–1.38) | **<0.001** | 1.31 (1.16–1.49) | **<0.001** |
| Insurance status at presentation | | | | |
| Uninsured (self-pay) | Base | | Base | |
| Private (BCBS, Commercial) | 1.09 (0.54–2.2) | 0.802 | 1.65 (0.71–3.83) | 0.240 |
| Public (Medicare, Medicaid, Tricare) | 0.76 (0.21–2.77) | 0.680 | 0.50 (0.09–2.21) | 0.325 |
| Time to presentation | | | | |
| Within 24 hrs | Base | | Base | |
| Within 48 hrs | 1.80 (0.48–6.57) | 0.387 | 1.84 (0.47–7.18) | 0.380 |
| Within 72 hrs | 0.57 (0.18–1.74) | 0.321 | 0.35 (0.09–1.43) | 0.145 |
| >72 hrs | 0.78 (0.37–1.65) | 0.520 | 0.64 (0.26–1.58) | 0.330 |

BCBS = Blue Cross Blue Shield; CI = confidence interval; OR = odds ratio; SVI = Social Vulnerability Index.
*Simple logistic regression. Boldface indicates statistical significance (<0.05).

Communities Index were associated with greater firearm-associated injuries and nonaccidental injuries. Consequently, patients having higher social vulnerability could be more prone to mechanisms of injury that are inherently more violent and severe, potentially leading to more devastating ocular trauma, hence poorer presenting VA. This increased risk may be attributed to various factors such as higher rates of community violence, systemic inequities, and limited access to health care as highlighted in our cohort where most patients lacked insurance (81.9%). Geographical location also played a role where we found greater distances to our eye center being significantly associated with higher odds of poor presenting VA on both univariable and multivariable (OR = 1.89, $P = 0.010$; OR = 2.55, $P = 0.002$, respectively). This finding could be attributed to longer distances causing delays in care, which result in more severe VAs at presentation. It may also reflect that patients traveling from more distant regions to our center tend to have more severe injuries, as less serious cases are typically managed locally by community physicians and more complex cases requiring specialized care being referred to tertiary centers. These observations emphasize how geographic and systemic disparities can contribute to poorer visual outcomes in vulnerable populations. Given these challenges, it is essential to examine not only the initial visual outcomes but also the long-term effects of social vulnerability and its impact on final visual outcomes over time.

Longitudinal changes in VA over time were not significantly affected by the SVI. Although patients with higher SVI had worse VA initially, their subsequent changes in VA followed a similar trend to those of less vulnerable populations. Several factors may explain this: first, in the acute setting, VA can be underestimated or overestimated due to pain, edema, and the presence of concomitant fractures, which can affect the initial assessment. Another potential explanation is the follow-up that all patients receive at our tertiary care center, ensuring equal access to care regardless of social vulnerability. Next, this could also be reflective of the follow-up protocols in place, which ensure consistent monitoring and management of patients, minimizing the impact of delays in initial care. Additionally, when examining insurance status, most patients (81.6%) initially presented to our eye center uninsured. However, this percentage decreased to 22.4% when followed over time, which may have contributed to the mitigation of the observed longitudinal effect of SVI. Furthermore, although geographic distance from the treatment center initially contributed to poorer presenting VA ($P = 0.002$), distance did not significantly correlate with the rate of decline in VA over time as reported by the multivariable analysis adjusted

Table 3. Univariable and Multivariable Analyses Assessing Predictive Factors of Rates of Visual Acuity Change over Time in Patients Presenting for Open Globe Injury

| Characteristic | Univariable Analysis | | Multivariable Analysis | |
|---|---|---|---|---|
| | *Coefficient (95% CI)* | P *Value** | *Coefficient (95% CI)* | P *Value** |
| SVI theme overall | | | | |
| Low vulnerability (0%–25%) | Base | | Base | |
| Moderate vulnerability (25%–50%) | −0.032 (−0.155 to 0.092) | 0.612 | 0.054 (−0.063 to 0.185) | 0.361 |
| High vulnerability (50%–75%) | −0.071 (−0.194 to 0.051) | 0.254 | 0.010 (−0.112 to 0.133) | 0.865 |
| Very high vulnerability (75%–100%) | −0.046 (−0.161 to 0.069) | 0.428 | 0.009 (−0.109 to 0.127) | 0.879 |
| Distance | | | | |
| >30 miles | 0.015 (−0.040 to 0.069) | 0.600 | −0.009 (−0.082 to 0.066) | 0.821 |
| Sex | | | | |
| Male | Base | | Base | |
| Female | 0.052 (−0.003 to 0.107) | 0.066 | 0.016 (−0.052 to 0.084) | 0.638 |
| Race | | | | |
| Non-Black | Base | | Base | |
| Black | 0.076 (0.020–0.132) | **0.008** | 0.103 (0.025–0.181) | **0.010** |
| Ethnicity | | | | |
| Non-Hispanic or Latino | Base | | Base | |
| Hispanic or Latino | −0.048 (−0.102 to 0.007) | 0.086 | 0.042 (−0.029 to 0.114) | 0.245 |
| Baseline age, per decade older | 0.020 (0.008–0.032) | **0.001** | 0.025 (0.011–0.040) | **<0.001** |
| Insurance status most used during follow-up | | | | |
| Uninsured (self-pay) | Base | | Base | |
| Private (BCBS, Commercial) | 0.048 (−0.013 to 0.108) | 0.120 | 0.038 (−0.040 to 0.115) | 0. 338 |
| Public (Medicare, Medicaid, Tricare) | 0.050 (−0.030 to 0.129) | 0.222 | −0.001 (−0.103 to 0.102) | 0.993 |
| Workers' compensation | −0.042 (−0.174 to 0.089) | 0.525 | −0.056 (−0.229 to 0.116) | 0.522 |
| WHO presenting VA severity | | | | |
| Mild | Base | | Base | |
| Moderate | −0.071 (−0.199 to 0.056) | 0.272 | −0.059 (−0.195 to 0.078) | 0.397 |
| Severe | −0.017 (−0.100 to 0.067) | 0.690 | −0.068 (−0.171 to 0.035) | 0.195 |
| Time from injury to presentation | | | | |
| Within 24 hrs | Base | | Base | |
| Within 48 hrs | 0.092 (−0.028 to 0.212) | 0.132 | 0.096 (−0.049 to 0.242) | 0.191 |
| Within 72 hrs | 0.076 (−0.034 to 0.185) | 0.175 | 0.074 (−0.077 to 0.225) | 0.338 |
| >72 hrs | 0.004 (−0.090 to 0.098) | 0.933 | −0.001 (−0.112 to 0.110) | 0.983 |

CI = confidence interval; BCBS = Blue Cross Blue Shield; SVI = Social Vulnerability Index; VA = visual acuity; WHO = World Health Organization.
*Linear mixed models. Boldface indicates statistical significance (<0.05).

for potential confounders ($P = 0.821$). As a result, distance could be acting more as an indicator for worse presenting injury severity rather than directly influencing the rate of VA decline over time. This highlights the complexity of OGI recovery and suggests that, while severe injuries result in immediate significant visual impairment, they may also result in a more stable long-term prognosis due to the absence of further significant declines.

Black race and older age were both significantly associated with poorer VA at presentation and a greater decline over time. However, only age remained a significant factor in worse outcomes at final VA on the multivariable analysis. In fact, final visual outcome was not influenced by either SVI or distance and other demographic characteristics except older baseline age at injury presentation ($P < 0.001$). This could be explained by older individuals generally having a reduced regenerative capacity of ocular tissues which may impair wound healing and functional recovery after trauma. In fact, the literature shows a limited recovery capacity and higher preexisting ocular pathologies in older patients negatively impact the prognosis of OGI.[16] Additionally, poor visual outcomes in older patients can be attributed to falls being the leading cause of OGIs in this population, frequently leading to rupture-type injuries rather than penetrating lacerations.[17,18] The literature has shown that ruptures are associated with larger wounds and lower baseline VA at presentation compared to lacerations, with a worse final visual outcomes.[19] The fact that neither SVI nor geographic distance significantly influenced the final visual outcome further supports our observation that the rates of VA decline over time were not affected by social vulnerability. While patients from more vulnerable backgrounds may start with worse VA, the quality of care and equitable access to treatment at our tertiary care center allow for improvements over time, mitigating the long-term effects of social vulnerability on visual outcomes. Furthermore, previous studies have suggested the SVI to serve as an indicator for long-term follow-up in chronic diseases like glaucoma.[10] Social Vulnerability Index may be particularly relevant in the management of chronic conditions, where long-term adherence to medication and follow-up visits are critical for preventing disease progression. However, its utility may be less pronounced in acute and severe conditions that immediately compromise

Table 4. Univariable and Multivariable Analysis Assessing Risk Factors for Poor Final Visual Acuity (Worse than 20/200) after Open Globe Injury

| Characteristic | Univariable Analysis | | Multivariable Analysis | |
|---|---|---|---|---|
| | *OR (95% CI)* | P *Value** | *OR (95% CI)* | P *Value** |
| SVI theme overall | | | | |
| Low vulnerability (0%–25%) | Base | | Base | |
| Moderate vulnerability (25%–50%) | 1.17 (0.45–3.05) | 0.740 | 1.64 (0.40–6.63) | 0.489 |
| High vulnerability (50%–75%) | 1.10 (0.46–2.64) | 0.824 | 1.71 (0.53–5.58) | 0.373 |
| Very high vulnerability (75%–100%) | 0.87 (0.39–1.97) | 0.745 | 1.72 (0.56–5.28) | 0.339 |
| Distance | | | | |
| >30 miles | 1.81 (1.16–2.83) | **0.009** | 1.38 (0.73–2.60) | 0.318 |
| Sex | | **0.010** | | 0.626 |
| Male | Base | | Base | |
| Female | 2.00 (1.18–3.38) | | 1.22 (0.55–2.66) | |
| Race | | **0.010** | | 0.085 |
| Non-Black | Base | | Base | |
| Black | 1.94 (1.17–3.21) | | 2.20 (0.90–5.41) | |
| Ethnicity | | **0.003** | | 0.780 |
| Non-Hispanic or Latino | Base | | Base | |
| Hispanic or Latino | 0.51 (0.33–0.79) | | 0.90 (0.42–1.92) | |
| Baseline age, per decade older | 1.39 (1.24–1.57) | **<0.001** | 1.36 (1.18–1.57) | **<0.001** |
| Most used insurance status during follow-up | | | | |
| Uninsured (self-pay) | Base | | Base | |
| Private (BCBS, Commercial) | 1.53 (0.88–2.66) | 0.129 | 1.18 (0.58–2.37) | 0.655 |
| Public (Medicare, Medicaid, Tricare) | 3.17 (1.50–6.69) | **0.002** | 1.67 (0.58–2.39) | 0.322 |
| Workers' compensation | 0.75 (0.30–1.89) | 0.543 | 1.02 (0.24–4.37) | 0.970 |
| WHO presenting VA severity | | | | |
| Mild | Base | | Base | |
| Moderate | 4.42 (0.82–23.86) | 0.084 | 4.82 (0.69–33.67) | 0.113 |
| Severe | 22.62 (5.19–98.46) | **<0.001** | 17.54 (3.43–89.83) | **0.001** |
| Time from injury to presentation | | | | |
| Within 24 hrs | Base | | | |
| Within 48 hrs | 1.03 (0.35–3.03) | 0.954 | 0.60 (0.16–2.30) | 0.454 |
| Within 72 hrs | 1.65 (0.53–5.19) | 0.391 | 2.85 (0.45–17.80) | 0.264 |
| >72 hrs | 0.64 (0.31–1.33) | 0.232 | 0.80 (0.31–2.08) | 0.647 |

BCBS = Blue Cross Blue Shield; CI = confidence interval; OR = odds ratio; SVI = Social Vulnerability Index; VA = visual acuity; WHO = World Health Organization.
*Simple logistic regression. Boldface indicates statistical significance (<0.05).

vision, such as OGI. In these acute cases, follow-up and recovery are often dictated by the urgency of care and the life-changing impact of acute vision loss, making social vulnerability less of an influence on these factors. Even though patients with higher social vulnerability may face greater systemic challenges, the immediate need for treatment and the severity of vision loss in the acute setting likely override these barriers. As a result, while SVI may be a crucial factor in long-term disease management like glaucoma, it may not significantly affect follow-up and recovery in acute ocular injuries. This hypothesis would most likely explain our findings, where SVI had a significant impact on worse presenting VA but did not influence rates of VA over time or final visual outcomes.

Our study offers a multifaceted approach to understanding the potential role of SVI in the context of OGI. It is based on a large cohort and serves as a dynamic tool, exploring the impact of SVI on VA and its evolution over time. Additionally, we accounted for various sociodemographic and geographical factors, providing a more comprehensive analysis of how these elements interact with recovery. Previous studies examining the relationship between the SVI and OGI have typically relied on simpler inferential statistics, such as Pearson correlation or chi-square tests, to report associations between SVI and presenting VA.[15] These methods provide valuable insights into the presence of relationships between variables but are limited in their ability to account for complex multilevel factors such as time or individual variability. In contrast, our study utilized linear mixed models, a more sophisticated approach that allows us to account for the temporal dynamics of visual recovery after OGI. By incorporating repeated measurements and modeling both fixed effects (such as the influence of SVI) and random effects (accounting for patient-level variability), we were able to capture how SVI influences VA changes over time. This approach provides a more comprehensive understanding of the longitudinal impact of social vulnerability on recovery, which simpler statistical methods cannot fully address. Additionally, by including sociodemographic and geographical factors, our analysis considers a broader range of influences on visual outcomes, offering a more nuanced perspective than studies that have relied on static, snapshot measurements of SVI. Insurance status, for example, is one

of several factors that changed over time that also influence VA outcomes. This highlights the dynamic nature of the covariables included in our study having a significant impact on recovery and overall visual outcomes.

However, several limitations of this study must be noted. No data were collected regarding whether these patients had ocular comorbidities, such as cataracts or glaucoma, or information on their ocular surgical history, both of which could have influenced the outcomes in terms of OGI and VA. While we recorded the VA at presentation, we lacked data on their VA before their initial visit to the ED, which would have been valuable to consider. Additionally, we did not have information on whether patients were initially evaluated by a local provider before being referred to our eye center, which could have influenced the severity of presentation and treatment delays. No data were collected on whether patients traveled by air or by car to reach the medical center. This is a significant factor as travel time can vary considerably depending on the mode of transportation. Additionally, our study used point-to-point distance, which does not account for variables such as road conditions, traffic congestion, and detours that can affect travel time. These factors may result in an underestimation or overestimation of actual travel times, potentially influencing the association between distance and visual outcomes. Furthermore, the follow-up period in this study ranged up to a maximum of 3 years, with an average of 1.4 years. While this duration provided valuable insights, some patients were lost to follow-up, which could have limited the ability to fully assess long-term visual outcomes. Future studies with extended follow-up periods would help to better capture the longitudinal changes in VA and further enhance our understanding of recovery trends over time. While we acknowledge that the logMAR scale is not a linear, we used a linear model to assess VA changes, consistent with previous studies in the literature.[14]

In conclusion, greater distances from the eye ED and higher social vulnerability were associated with a higher risk of worse VA at presentation. However, no significant impact of SVI or distance was observed on long-term VA changes and final visual outcome. While improving health care access for vulnerable populations may not directly enhance long-term visual outcomes, it remains crucial for reducing disparities by ensuring earlier presentation and timely intervention.

## Footnotes and Disclosures

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Abbreviations and Acronyms:
**CI** = confidence interval; **ED** = emergency department; **logMAR** = logarithm of the minimum angle of resolution; **OGI** = open globe injury; **OR** = odds ratio; **SVI** = Social Vulnerability Index; **VA** = visual acuity.

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